

Our Digital World — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper. Every sum is built to work by hand, using small numbers or powers of ten.
- A label alone is rarely a full answer: where a question says **explain** or **describe**, give the action or the reason. A defence only scores with its reason — “install a firewall” earns nothing; say what it does that stops the attack.
- The mark for each question is shown in square brackets on the right.
- **Fact-check checklist:** 1 Source — is it named, and can you check who they are? 2 Date — is there one, and does it fit the event? 3 Evidence — can you trace a link, the data or the full quote? 4 Hurry — is it pushing you to react or forward it? 5 Other outlets — do reliable, independent outlets report the same thing?
- **What each defence does:** a **firewall** blocks network traffic that breaks its rules; **antivirus** scans files and programs against known malware; **anti-spyware** detects and removes programs that secretly watch you.

SECTION 1 — WORDS AND TOOLS

1. Four things happened to Diya yesterday. Write **A** (active) or **P** (passive) in the box [4]
to show which part of her digital footprint each one adds to.

What happened	A or P
She posted a photo of her rangoli on Instagram	
Her cousin tagged her in a video of the family dinner	
A shopping app recorded which pages she looked at	
She filled in a sign-up form for a cricket coaching camp	

2. Write the right word in each box: *catfishing*, *trolling*, *fraping* or *dissing*. [4]

What the person does	The word
Posts “you’re a joke, just quit” under strangers’ videos every day, to upset them	
Sends an embarrassing photo of a classmate to a group chat of forty, with a mocking caption	
Builds an account from stolen photos and an invented life, and chats to a stranger for weeks	
Posts as a friend from the friend’s phone, which was left logged in	

3. For each description, say whether it is malware, and name it. Choose from *virus*, *spyware*, *ransomware*, *brute force attack*. A row scores only if **both** boxes are right. [4]

What it does	Malware? (Yes / No)	Name
Runs unseen and sends a copy of every key you press to a stranger		
A program tries 10,000 passwords one after another against a login until one works		
Encrypts your photos and documents, then demands ₹5,000 for the key		
Hides inside a genuine file and copies itself into other files when the file is shared		

SECTION 2 – LOOK AT THE EVIDENCE

4. Read the forward below and the result of a quick search.

WHATSAPP FORWARD • “FORWARDED MANY TIMES”	
THE POST	*URGENT* Scientists in Germany have PROVED that Class 7 students who eat 2 almonds before an exam score 20 marks higher. No tuition needed. Teachers do not want you to know this. Forward to 10 groups TODAY or you will lose marks!
QUICK SEARCH	The first sentence appears word for word on dozens of pages. No scientist, university or study is named on any of them.

- (a) Name **two** of the five checks that this post fails. For each one, quote the words from the post that show it. [2]

- (b) Aarav says: “Ten of my friends have already forwarded it, so it must be true.” [1]
Explain why that does not make it true.

5. Meera finds this photograph and wants to use it in her school project.

LICENCE NOTICE · SHOWN UNDER A PHOTOGRAPH ON A PHOTO-SHARING SITE	
TITLE	Monsoon at Cubbon Park
CREATOR	Kavya Shetty
SOURCE	photoshare.example/kavya-s
LICENCE	Creative Commons BY-NC

(a) Write the credit line Meera must put under the photograph. [2]

(b) A friend wants to print the photograph on T-shirts and sell them for ₹300 each. [2]
He says: “It’s fine — we’ll credit Kavya on every one.” State whether this is allowed, and explain using the licence.

6. Name the guard — *firewall*, *antivirus* or *anti-spyware* — that does each job. [3]

The job	Guard
A school stops every computer on its network connecting to a gambling website	
A downloaded file is compared with a database of known malware and moved to quarantine	
A free emoji app that secretly records what you type is found and removed	

7. A phone lock uses a PIN made only of digits. An attacker’s program makes **1,000** [4]
guesses every second. Complete the table. Time to try them all = combinations
÷ 1,000.

Digits in the PIN	Combinations	Time to try them all
4	10,000	10 seconds
5		
6		

Write each number of combinations as a power of ten first.

8. Karan posts an angry comment, then deletes it two minutes later. A classmate had [2]
already taken a screenshot. Explain why the comment can still be found.

9. Neha is in a class group chat. She sees several classmates mocking another girl’s [2]
photo. Neha is not one of them. Describe **two** things Neha should do, each with a
reason.

End of paper. Check that every defence you named comes with its reason, and every calculation with its working.